



EXAM PAPERS PRACTICE

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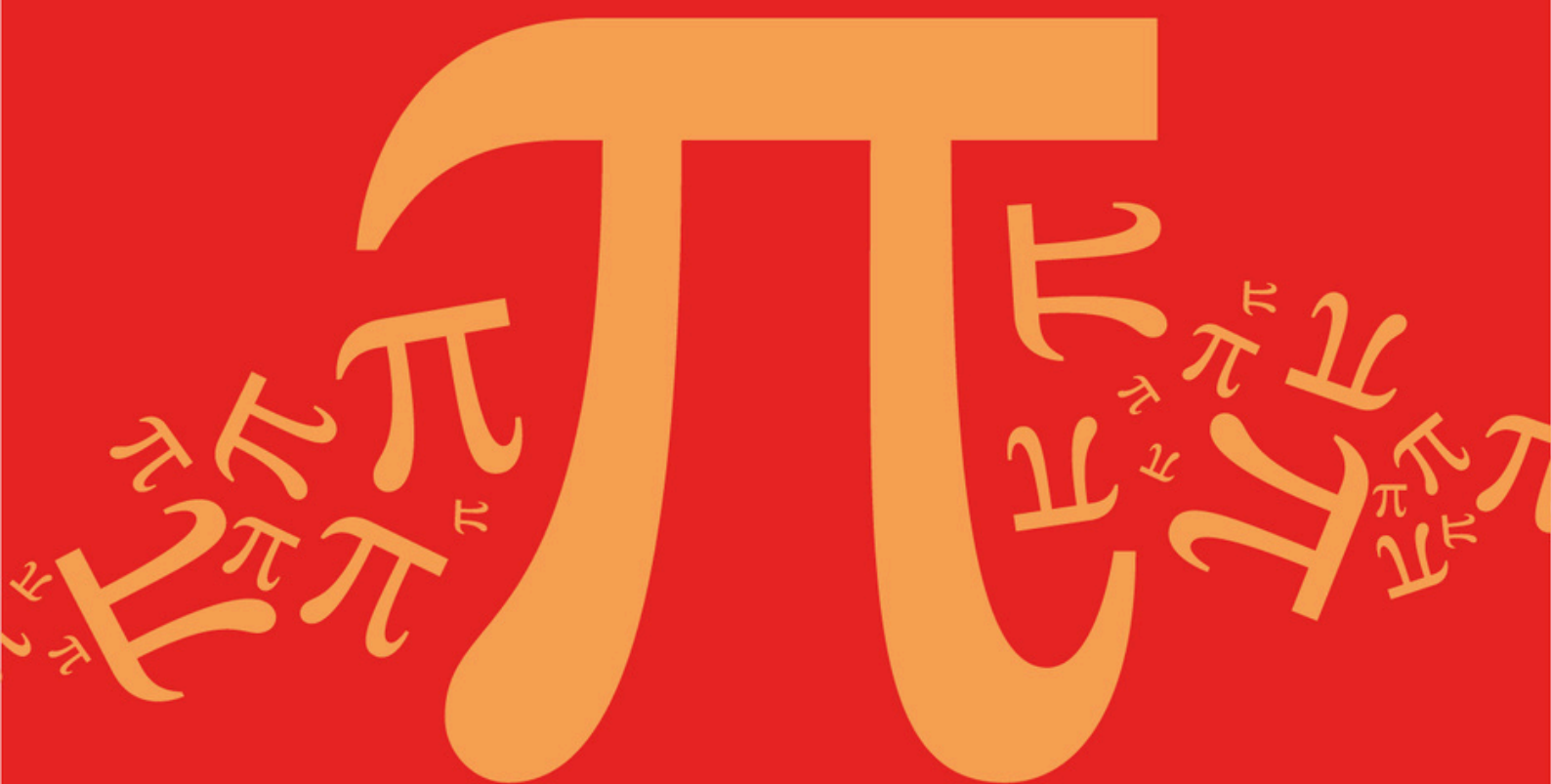
Practice questions created by actual examiners and assessment experts

Detailed mark scheme

Suitable for all boards

Designed to test your ability and thoroughly prepare you

AQA A Level Further Mathematics (7367)



Topic

D: Further Algebra and Functions

Sub topic

D4: Method of Differences

Mark Scheme

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Mark Scheme

Topic	D: Further Algebra and Functions
Sub topic	D4: Method of Differences

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**To be used by all students and teachers for
AQA A Level Further Mathematics (7367)
for exams from 2025 onwards.**

Students and teachers of other boards may also find this useful

Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Shows the result is true with at least one intermediate step.	AO1.1b	B1	$\frac{1}{r+2} - \frac{1}{r+3} = \frac{r+3-(r+2)}{(r+2)(r+3)}$ $= \frac{1}{(r+2)(r+3)}$ $\sum_{r=1}^n \left(\frac{1}{(r+2)(r+3)} \right) = \sum_{r=1}^n \left(\frac{1}{r+2} - \frac{1}{r+3} \right)$ $= \left(\frac{1}{3} - \frac{1}{4} \right) + \left(\frac{1}{4} - \frac{1}{5} \right) + \left(\frac{1}{5} - \frac{1}{6} \right) + \dots$ $+ \left(\frac{1}{n+1} - \frac{1}{n+2} \right) + \left(\frac{1}{n+2} - \frac{1}{n+3} \right)$ $= \frac{1}{3} - \frac{1}{n+3}$ $= \frac{n+3-3}{3(n+3)}$ $= \frac{n}{3(n+3)}$
(b)	Writes at least three corresponding terms of $\frac{1}{r+2}$ and $\frac{1}{r+3}$ Must include the 1 st and n^{th} terms and at least the 2 nd term or the $(n-1)^{\text{th}}$ term.	AO1.1a	M1	
	Correctly uses the method of differences to reduce the sum to two terms.	AO1.1b	A1	
	Completes fully correct proof to reach the required result. This mark is only available if all previous marks have been awarded.	AO2.1	R1	
				AG
				Total 4 marks

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Q2.

(a)
$$\frac{1}{5r-2} - \frac{1}{5r+3} = \frac{5r+3-(5r-2)}{(5r-2)(5r+3)}$$

condone omission of brackets for M1

M1

$$= \frac{5}{(5r-2)(5r+3)}$$

$$A = 5$$

A1cso

2

- (b) Attempt to use method of differences
at least 2 terms of correct form seen

M1

$$k \left\{ \frac{1}{3} - \frac{1}{5n+3} \right\}$$

correct cancellation leaving correct two fractions

A1

$$k \left\{ \frac{(5n+3)-3}{3(5n+3)} \right\}$$

attempt to write with common denominator

m1

$$S_n = \frac{1}{5} \left\{ \frac{(5n+3)-3}{3(5n+3)} \right\} = \frac{n}{3(5n+3)}$$

AG $k = \frac{1}{5}$ used correctly throughout

A1cso

4

(c) $S_{\infty} = \frac{1}{15}$

B1

1

[7]

Q3.

(a) $f(r) - f(r-1) = r^2(2r^2-1) - (r-1)^2(2(r-1)^2-1)$

condone one slip here

$$= 2r^4 - r^2 - (r^2 - 2r + 1)(2r^2 - 4r + 1)$$

attempt to multiply out "their" $f(r-1)$

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M1

$$= 2r^4 - r^2 - (2r^4 - 8r^3 + 11r^2 - 6r + 1)$$

$f(r)$ & $f(r-1)$ expanded correctly condone correct unsimplified

A1

$$= 8r^3 - 12r^2 + 6r - 1 = (2r-1)^3$$

AG

A1cso

3

(b) Attempt to use method of differences

M1

$$f(2n) - f(n)$$

$$(2n)^2\{2(2n^2) - 1\} - n^2(2n^2 - 1)$$

m1

$$f(2n) - f(n) = 4n^2 (8n^2 - 1) - n^2 (2n^2 - 1)$$

A1

$$= 30n^4 - 3n^2$$

$$= 3n^2 (10n^2 - 1)$$

AG be convinced

A1cso

4

[7]

Q4.

- (a) Attempt to put LHS over common denominator

$$\frac{2^{r+1}(r+1) - 2^r(r+2)}{(r+1)(r+2)}$$

any form

M1

$$= \frac{r(2^{r+1} - 2^r)}{(r+1)(r+2)} = \frac{r2^r}{(r+1)(r+2)} \text{ must see } r2^{r+1} = 2r2^r$$

clearly shown as AG

A1

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A1

3

- (b) $\frac{2^2}{3} - \frac{2}{2}$ ©2025 Exam Papers Practice. All rights reserved.

$$\frac{2^3}{4} - \frac{2^2}{3}$$

.....

3 rows indicated (PI)

M1

$$\frac{2^{31}}{32} - \frac{2^{30}}{31}$$

$$S_{30} = \frac{2^{31}}{32} - 1 \text{ or } S_n = \frac{2^{n+1}}{n+2} - 1$$

A1

$$= 2^{26} - 1$$

CAO

A1

3

[6]

Q5.

(a) $u_r - u_{r-1} = \frac{1}{6}r(r+1)(4r+11) - \frac{1}{6}(r-1)r(4r+7)$

M1

Correct expansion in any form, eg $\frac{1}{6}r(4r^2 + 15r + 11 - 4r^2 - 3r + 7)$

A1

$= r(2r + 3)$

AG

A1

3

(b) Attempt to use method of differences

M1

$S_{100} = u_{100} - u_0$

A1

$= 691850$

CAO

A1

3

[6]

Q6.

(a) $(r+1)! = (r+1)r(r-1)!$

M1

Result

AG

A1

2

(b) **Attempt** to use method of differences

M1

$\sum_{r=1}^n (r^2 + r - 1)(r-1)! = (n+1)! + n! - 1! - 0!$

A1

$$(n+1)! = (n+1)n!$$

Must be seen

m1

$$(n+2)n! - 2$$

AG

A1

4

[6]

Q7.

(a)

$$f(r) - f(r-1)$$

$$= \frac{1}{4}r^2(r+1)^2 - \frac{1}{4}(r-1)^2r^2$$

M1

$$= \frac{1}{4}r^2(r^2 + 2r + 1 - r^2 + 2r - 1)$$

Correct expansions of $(r+1)^2$ and $(r-1)^2$

A1

$$= r^3$$

AG

A1

3

(b)

$$r = n: \quad n^3 = \frac{1}{4}n^2(n+1)^2 - \frac{1}{4}(n-1)^2n^2$$

For either $r = n$ or $r = 2n$. PI

M1A1

$$r = 2n:$$

$$(2n)^3 = \frac{1}{4}(2n)^2(2n+1)^2 - \frac{1}{4}(n-1)^2(2n)^2$$

A1

$$\sum_{r=n}^{2n} r^3 = \frac{1}{4} \cdot 4n^2(2n+1)^2 - \frac{1}{4}(n-1)^2n^2$$

M1

$$= \frac{3}{4} n^2 (5n + 1)(n + 1)$$

AG

A1

Alternatively

$$\sum_{r=1}^{2n} r^3 \text{ and } \sum_{r=1}^{n-1} r^3$$

stated M1A1A1

(M1 for either)

Difference

M1

Answer

A1

5

[8]

Q8.

- (a) Attempt to expand $(2r + 1)^3 - (2r - 1)^3$

M1

$(2r + 1)^3$ or $(2r - 1)^3$ expanded

A1

$$24r^3 + 2$$

AG

A1

3

- (b) $r = 1 \quad 3^3 - 1^3 = 24 \times 1^2 + 2$

$$r = 2 \quad 5^3 - 3^3 = 24 \times 2^2 + 2$$

3 rows seen

Do not allow M1 for $(2n + 1)^3 - 1$ not equal to anything

M1A1

$$r = n \quad (2n + 1)^3 - (2n - 1)^3 = 24 \times n^2 + 2$$

$$(2n + 1)^3 - 1 = 24 \sum_{r=1}^n r^2 + 2n$$

A1

$$8n^3 + 12n^2 + 6n + 1 - 1 - 2n = 24 \sum_{r=1}^n r^2$$

M1 for multiplication of bracket or taking $(2n + 1)$ out as a factor

M1

$$8n^3 + 12n^2 + 4n = 24 \sum_{r=1}^n r^2$$

CAO

A1

$$\sum_{r=1}^n r^2 = \frac{1}{6} n(n+1)(2n+1)$$

AG

A1

6

[9]

Q9.

(a) $\tan((r+1)x - rx) = \frac{\tan(r+1)x - \tan rx}{1 + \tan(r+1)x \tan rx}$

M1A1

Multiplying up

A1

Printed result

AG

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A1

4

(b) $x = \frac{\pi}{50}$

$$\tan \frac{\pi}{50} \tan \frac{2\pi}{50} = \frac{\tan \frac{2\pi}{50} - \tan \frac{\pi}{50}}{\tan \frac{\pi}{50} \tan \frac{2\pi}{50}} - 1$$

$$\tan \frac{2\pi}{50} \tan \frac{3\pi}{50} = \frac{\tan \frac{3\pi}{50} - \tan \frac{2\pi}{50}}{\tan \frac{\pi}{50} \tan \frac{3\pi}{50}} - 1$$

At least three lines to be shown

Accept if x's used

M1A1

.....

$$\tan \frac{19\pi}{50} \tan \frac{20\pi}{50} = \frac{\tan \frac{20\pi}{50}}{\tan \frac{\pi}{50}} - \frac{\tan \frac{19\pi}{50}}{\tan \frac{\pi}{50}} - 1$$

Clear cancellation

m1

$$\text{Sum} = \frac{\tan \frac{20\pi}{50}}{\tan \frac{\pi}{50}} - \frac{\tan \frac{\pi}{50}}{\tan \frac{\pi}{50}} - 19$$

Sum

$$= \frac{\tan \frac{2\pi}{5}}{\tan \frac{\pi}{50}} - 20$$

AG

A1

A1

5

[9]

Q10.

(a) $f(r+1) - f(r) = r(r+1)^2 - (r-1)r^2$

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M1

$$= r(r^2 + 2r + 1 - r^2 + r)$$

any expanded form

A1

$$= r(3r+1)$$

AG

OE

A1

3

(b)

$$\left. \begin{array}{l} r=50 \\ r=51 \\ r=99 \end{array} \right\} \begin{array}{l} f(51) - f(50) \\ f(52) - f(51) \\ f(100) - f(99) \end{array} \Bigg\} \text{PI}$$

clearly shown. Accept $\sum_{r=50}^{99} - \sum_{r=50}^{49}$

M1A1

$$\sum_{r=50}^{99} r(3r+1) = f(100) - f(50)$$

clear cancellation

m1

$$= 867500$$

cao

A1F

4

[7]

Q11.

(a) $\frac{1}{r^2} - \frac{1}{(r+1)^2} = \frac{(r+1)^2 - r^2}{r^2(r+1)^2}$

M1

$$= \frac{2r+1}{r^2(r+1)^2}$$

AG

A1

2

(b) $\frac{3}{1^2 \times 2^2} = \frac{1}{1^2} - \frac{1}{2^2}$

$$\frac{5}{2^2 \times 3^2} = \frac{1}{2^2} - \frac{1}{3^2}$$

$$\frac{7}{3^2 \times 4^2} = \frac{1}{3^2} - \frac{1}{4^2}$$

A1 for at least 3 lines

M1A1

$$\frac{2n+1}{n^2(n+1)^2} = \frac{1}{n^2} - \frac{1}{(n+1)^2}$$

Clear cancellation

M1

$$1 - \frac{1}{(n+1)^2}$$

A1F

4

[6]

Examiner reports

Q1.

Part (a) was well answered. A common error, however, was the incorrect use of brackets or missing brackets, for example $r + 3 - r + 2 = 1$.

Part (b) was also well answered. Many of those who failed to successfully prove the required identity wrote the list of differences beyond $r = n$.

Q2.

- (a) Most students found the correct value of A , but others, usually those who omitted brackets, thought that A was equal to 1. A few students assumed that the numerator was A and simply cross multiplied to find the value of A , but this approach did not score full marks.
- (b) The method of differences was generally understood. The printed answer no doubt helped many to amend their solution when they realised that they needed to divide their sum by 5 in order to obtain the given answer. Some did this legitimately and scored full marks; others used dubious reasoning and lost at least the final mark.
- (c) The basic idea of a limit, as used in MFP1, was required here in order to find the value of the sum as n **tended to infinity**. A common wrong answer was $\frac{1}{3}$ and was often as a result of the earlier working where the factor of $\frac{1}{5}$ had been omitted. Other common wrong answers were 1, 0 and infinity.

Q3.

- (a) This part of the question required stamina in expanding then simplifying the expression for $f(r) - f(r - 1)$. Some looked in vain for a common factor such as $(2r - 1)$. Those who made careful use of brackets and who set out their solution neatly were usually successful in showing that the resulting expression was equivalent to $(2r - 1)^3$.
- (b) The method of differences was generally used effectively. Some realised that the sum reduced to $f(2n) - f(n)$. Others used the method of differences on two separate series and also obtained $(2n)^2 (2(2n)^2 - 1) - n^2 (2n^2 - 1)$. Quite a few students did not use brackets and failed to obtain this precise expression. This prevented some from obtaining the printed answer without fudging their work.

Q4.

This question was well answered. In part (a), errors, when they occurred, were a lack of demonstration that $r2^{r+1} - r2^r$ was in fact equal to $r2^r$. And in part (b), errors included students writing a correct but unfinished answer in the sense that a correct answer had been obtained, but had been left in a form different that that required by the question.

Q5.

The algebra of part (a) of this question was well done, and many candidates were

successful with part (b). In part (b), however, in spite of the question stating 'Hence', some candidates chose to use alternative methods, usually using $\sum r$ and $\sum r^2$ from the formulae booklet; no credit was given for such attempts.

Q6.

Not all candidates were confident in their use of factorials, but most managed part (a). In part (b), however, although candidates understood the method of differences, quite a number failed to reach the final answer. It was common to see $0!$ written down as zero whilst others abbreviated their working, writing $n! + (n+1)!$ as $(n+2)n!$ without any intermediate step. As the answer was provided, this solution did not secure full marks.

Q7.

There were many good and completely correct solutions to this question. Virtually all candidates completed part (a) correctly, and in part (b), if an error occurred, it was usually

in the selection of an incorrect value for r at one end. For instance, $\sum_{r=n}^{2n} r^3$ was taken to be $f(2n) - f(n)$ rather than $f(2n) - f(n-1)$.

Q8.

Again part (a) was answered well, but solutions to part (b) were mixed. Generally speaking, the best solutions came from candidates who rewrote part (a) as $r^2 =$

$\frac{1}{24} \{ (2r+1)^3 - (2r-1)^3 - 2 \}$ before making their summation. Those candidates who preferred to use part (a) in the form in which it was printed either forgot to sum the 2's to make $2n$ or only partially divided by 24. A small number of candidates used the method of induction either through confusing the two methods of summation or by deliberately choosing an alternative method. Either way, no credit could be given.

Q9.

This question was surprisingly well done and attracted many completely correct solutions. When errors occurred, it was usually in the summation of terms in part (b). Candidates summed 20 terms instead of 19 which in turn led to some faking in arriving at the printed answer, especially the -20 . For instance it was not uncommon to see the summation

$$\frac{\tan \frac{21\pi}{50} - \tan \frac{\pi}{50}}{\tan \frac{\pi}{50}} - 20$$

written as

followed by the correct answer.

Q10.

Almost all candidates were successful with part (a) However, in part (b) a number of

candidates used $\sum r^2$ and $\sum r$ to evaluate $\sum_{50}^{99} r(3r+1)$ contrary to the requirement of the question and so, even with a correct answer, scored no marks. The most successful candidates for this part of the question were those who carefully wrote out a number of rows including the first and last row, to illustrate the cancellations. Some candidates went awry when writing down the first or last terms of the series.

Q11.

There were many fully correct answers to this question. A few candidates did not spot the connection between the parts (a) and (b). Otherwise, the only errors in part (b) were errors

of sign leading to the answer $1 + \frac{1}{(n+1)^2}$ or the summation of $n+1$ terms rather than n terms of the given series.



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